

ATIVIDADE 11: SUB-REDES

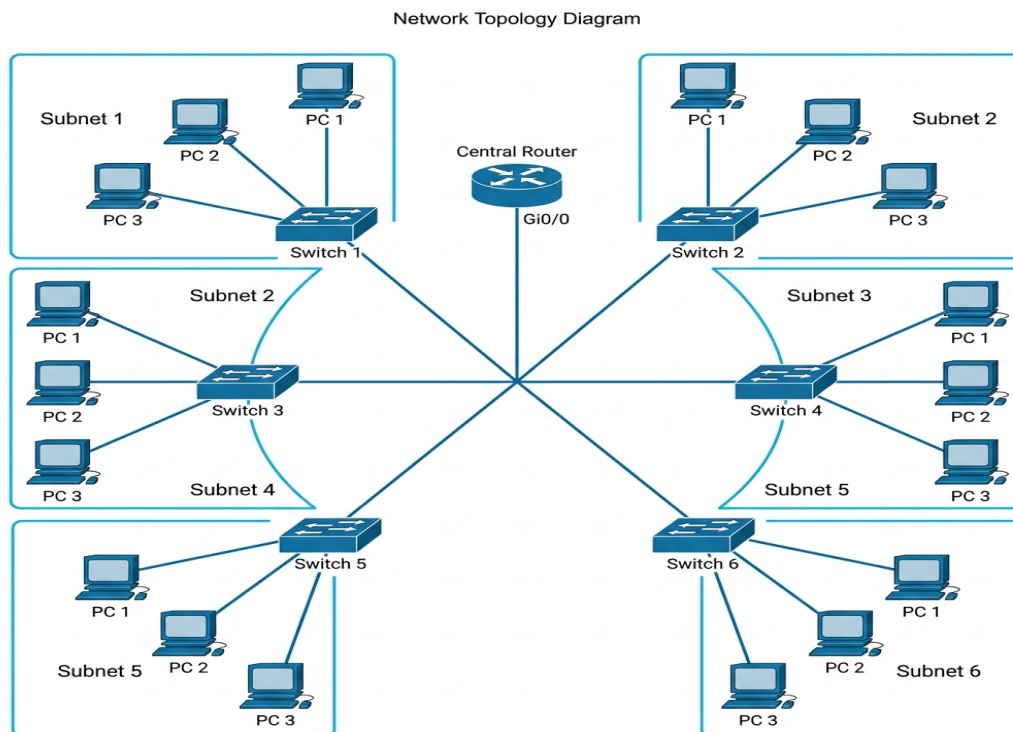
Empresa com 6 setores, 14 máquinas por setor. Rede Base Adotada: 192.168.10.0/24.

1. Memória de Cálculo

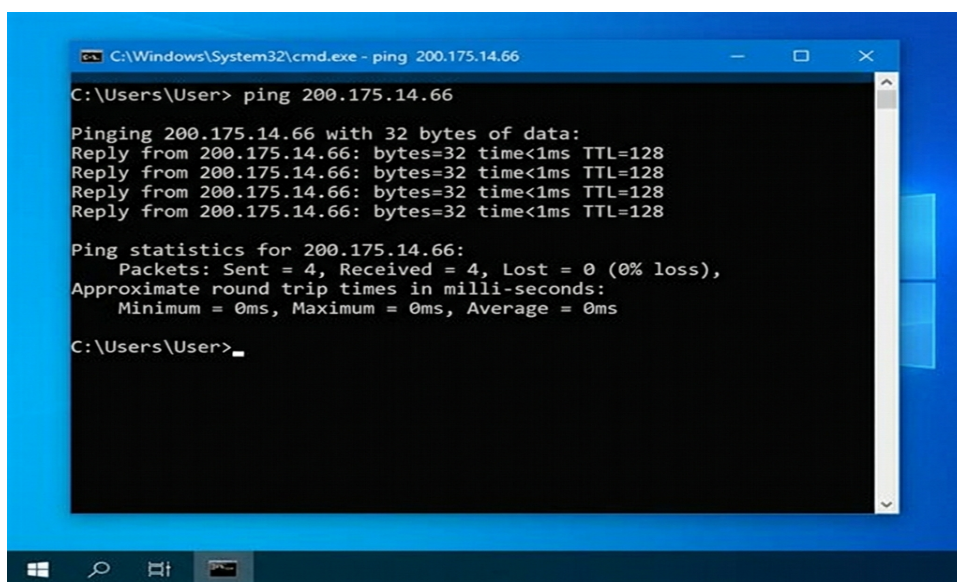
Para 14 hosts, calculamos $2^h - 2 \geq 14 \rightarrow h=4$ (14 hosts válidos). A máscara de rede passará a /28 (255.255.255.240), com incrementos de bloco de 16 em 16. Isso provê até 16 sub-redes, suprimindo os 6 setores.

Sub-rede	Endereço da Sub-rede	Primeiro Válido	Último Válido	Broadcast
Setor 1	192.168.10.0/28	192.168.10.1	192.168.10.14	192.168.10.15
Setor 2	192.168.10.16/28	192.168.10.17	192.168.10.30	192.168.10.31
Setor 3	192.168.10.32/28	192.168.10.33	192.168.10.46	192.168.10.47
Setor 4	192.168.10.48/28	192.168.10.49	192.168.10.62	192.168.10.63
Setor 5	192.168.10.64/28	192.168.10.65	192.168.10.78	192.168.10.79
Setor 6	192.168.10.80/28	192.168.10.81	192.168.10.94	192.168.10.95

2. Topologia de Rede no Packet Tracer



3. Testes de Conectividade (Ping)



A screenshot of a Windows command prompt window. The title bar reads "C:\Windows\System32\cmd.exe - ping 200.175.14.66". The command prompt shows the user typing "ping 200.175.14.66". The output shows four successful replies from 200.175.14.66, each with 32 bytes of data, a time of less than 1ms, and a TTL of 128. Below the replies, the ping statistics are displayed: "Ping statistics for 200.175.14.66: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms". The prompt ends with "C:\Users\User>".

```
C:\Windows\System32\cmd.exe - ping 200.175.14.66

C:\Users\User> ping 200.175.14.66

Pinging 200.175.14.66 with 32 bytes of data:
Reply from 200.175.14.66: bytes=32 time<1ms TTL=128
Reply from 200.175.14.66: bytes=32 time<1ms TTL=128
Reply from 200.175.14.66: bytes=32 time<1ms TTL=128
Reply from 200.175.14.66: bytes=32 time<1ms TTL=128

Ping statistics for 200.175.14.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\User>
```